

Computer Tools

DSA:8420 Advanced Math Programming

Optimization Computation

Layers of Optimization Computation:

- **Algorithms:** Simplex, Interior Point, Cutting Plane, Newton, ...
- **Solver/Package:**
 - ▶ Open-source: GLPK, ICOPT, SCIP, CBC, SeDuMi, ...
 - ▶ Commercial: Gurobi, Cplex, Mosek, COPT, BARON, ...
- **Modeling Language:** Pyomo, AMPL, CVX, GAMS, ...
- **Programming Language:** Python, C, Java, Julia, Matlab, ...

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In this video: Python $\rightarrow$ Pyomo $\rightarrow$ GLPK

Two More Layers

In this video:

Anaconda → Jupyter Notebook → Python → Pyomo → GLPK

- **GLPK**: GNU Linear Programming Kit
- **Pyomo**: Python Optimization Modeling Objects
- **Python**: a high-level, general-purpose programming language
- **Jupyter Notebook**: a web-based interactive computing notebook environment
- **Anaconda**: a distribution of the Python and R programming languages for scientific computing, aiming to simplify package management and deployment

Installation

- Download the Anaconda Distribution:
<https://www.anaconda.com/products/distribution>.
- Install Anaconda3

For Windows, open the Anaconda Command Prompt.
For Mac, open the Terminal.

- Create an environment in Anaconda
`conda create -n opt.in.python python`
- Activate the environment
`conda activate opt.in.python`

Installation

- Install Pyomo
`conda install -c conda-forge pyomo`
- Install GLPK
`conda install -c conda-forge glpk`
- Install and run Jupyter Notebook
`conda install jupyter`
`jupyter notebook`

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To run Jupyter Notebook after initial installation:

```
conda activate opt.in.python
jupyter notebook
```

Resources

- Detailed Pyomo workshop slides can be found at <http://www.pyomo.org/documentation>.
- An introduction to the Jupyter Notebook can be found at <https://jupyter-notebook.readthedocs.io/en/latest/notebook.html>
- Trouble shooting: “The installation failed” message when running a .pkg installer on OSX.
<https://docs.anaconda.com/anaconda/user-guide/troubleshooting/#the-installation-failed-message-when-running-a-pkg-installer-on-osx>